

```
service apache2 start
```

```
└─(muhammad@virus)-[~/TESTS]
```

```
└─$ sudo systemctl status apache2
```

● apache2.service - The Apache HTTP Server

Loaded: loaded (/usr/lib/systemd/system/apache2.service; disabled; preset: disabled)

Active: active (running) since Wed 2025-08-06 06:30:29 EDT; 4min 7s ago

Invocation: c9ba30b2d5a54b159cb8ae766c981e36

Docs: <https://httpd.apache.org/docs/2.4/>

Process: 29723 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)

Main PID: 29739 (apache2)

Tasks: 8 (limit: 19040)

Memory: 22.7M (peak: 22.9M)

CPU: 233ms

CGroup: /system.slice/apache2.service

└─29739 /usr/sbin/apache2 -k start

└─29742 /usr/sbin/apache2 -k start

└─29743 /usr/sbin/apache2 -k start

└─29744 /usr/sbin/apache2 -k start

└─29745 /usr/sbin/apache2 -k start

└─29746 /usr/sbin/apache2 -k start

└─29747 /usr/sbin/apache2 -k start

└─31076 /usr/sbin/apache2 -k start

Aug 06 06:30:29 virus systemd[1]: Starting apache2.service - The Apache HTTP Server...

Aug 06 06:30:29 virus systemd[1]: Started apache2.service - The Apache HTTP Server.

```
└─$ apache2 -v
```

Server version: Apache/2.4.65 (Debian)

Server built: 2025-07-29T17:52:31

```
└─(muhammad@virus)-[~/TESTS]
```

```
└─$ sudo nmap -sS -sV -T4 localhost
```

Starting Nmap 7.95 (<https://nmap.org>) at 2025-08-06 06:37 EDT

Nmap scan report for localhost (127.0.0.1)

Host is up (0.0000050s latency).

Other addresses for localhost (not scanned): ::1

Not shown: 998 closed tcp ports (reset)

PORT	STATE	SERVICE	VERSION
------	-------	---------	---------

80/tcp	open	http	Apache httpd 2.4.65 ((Debian))
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5432/tcp	open	postgresql	PostgreSQL DB 15.2 - 15.4
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Service detection performed. Please report any incorrect results at <https://nmap.org/submit/> .

Nmap done: 1 IP address (1 host up) scanned in 6.68 seconds

```
└─$ cat /etc/apache2/apache2.conf
```

```
# This is the main Apache server configuration file. It contains the
# configuration directives that give the server its instructions.
# See http://httpd.apache.org/docs/2.4/ for detailed information about
# the directives and /usr/share/doc/apache2/README.Debian about Debian specific
# hints.
#
#
# Summary of how the Apache 2 configuration works in Debian:
# The Apache 2 web server configuration in Debian is quite different to
# upstream's suggested way to configure the web server. This is because Debian's
# default Apache2 installation attempts to make adding and removing modules,
# virtual hosts, and extra configuration directives as flexible as possible, in
# order to make automating the changes and administering the server as easy as
# possible.
#
# It is split into several files forming the configuration hierarchy outlined
# below, all located in the /etc/apache2/ directory:
#
#   /etc/apache2/
#   |-- apache2.conf
#   |   `-- ports.conf
#   |-- mods-enabled
#   |   |-- *.load
#   |   `-- *.conf
#   |-- conf-enabled
#   |   `-- *.conf
#   `-- sites-enabled
```

```
#      `-- *.conf
#
#
# * apache2.conf is the main configuration file (this file). It puts the pieces
# together by including all remaining configuration files when starting up the
# web server.
#
# * ports.conf is always included from the main configuration file. It is
# supposed to determine listening ports for incoming connections which can be
# customized anytime.
#
# * Configuration files in the mods-enabled/, conf-enabled/ and sites-enabled/
# directories contain particular configuration snippets which manage modules,
# global configuration fragments, or virtual host configurations,
# respectively.
#
# They are activated by symlinking available configuration files from their
# respective *-available/ counterparts. These should be managed by using our
# helpers a2enmod/a2dismod, a2ensite/a2dissite and a2enconf/a2disconf. See
# their respective man pages for detailed information.
#
# * The binary is called apache2. Due to the use of environment variables, in
# the default configuration, apache2 needs to be started/stopped with
# /etc/init.d/apache2 or apache2ctl. Calling /usr/bin/apache2 directly will not
# work with the default configuration.

# Global configuration
```

#

#

ServerRoot: The top of the directory tree under which the server's
configuration, error, and log files are kept.

#

NOTE! If you intend to place this on an NFS (or otherwise network)
mounted filesystem then please read the Mutex documentation (available
at <URL:http://httpd.apache.org/docs/2.4/mod/core.html#mutex>);
you will save yourself a lot of trouble.

#

Do NOT add a slash at the end of the directory path.

#

#ServerRoot "/etc/apache2"

#

The accept serialization lock file MUST BE STORED ON A LOCAL DISK.

#

#Mutex file:\${APACHE_LOCK_DIR} default

#

The directory where shm and other runtime files will be stored.

#

DefaultRuntimeDir \${APACHE_RUN_DIR}

#

PidFile: The file in which the server should record its process

identification number when it starts.

This needs to be set in /etc/apache2/envvars

#

PidFile \${APACHE_PID_FILE}

#

Timeout: The number of seconds before receives and sends time out.

#

Timeout 300

#

KeepAlive: Whether or not to allow persistent connections (more than

one request per connection). Set to "Off" to deactivate.

#

KeepAlive On

#

MaxKeepAliveRequests: The maximum number of requests to allow

during a persistent connection. Set to 0 to allow an unlimited amount.

We recommend you leave this number high, for maximum performance.

#

MaxKeepAliveRequests 100

#

KeepAliveTimeout: Number of seconds to wait for the next request from the

same client on the same connection.

#

KeepAliveTimeout 5

These need to be set in /etc/apache2/envvars

User \${APACHE_RUN_USER}

Group \${APACHE_RUN_GROUP}

#

HostnameLookups: Log the names of clients or just their IP addresses

e.g., www.apache.org (on) or 204.62.129.132 (off).

The default is off because it'd be overall better for the net if people

had to knowingly turn this feature on, since enabling it means that

each client request will result in AT LEAST one lookup request to the

nameserver.

#

HostnameLookups Off

ErrorLog: The location of the error log file.

If you do not specify an ErrorLog directive within a <VirtualHost>

container, error messages relating to that virtual host will be

logged here. If you *do* define an error logfile for a <VirtualHost>

container, that host's errors will be logged there and not here.

#

ErrorLog \${APACHE_LOG_DIR}/error.log

#

LogLevel: Control the severity of messages logged to the error_log.

Available values: trace8, ..., trace1, debug, info, notice, warn,

error, crit, alert, emerg.

It is also possible to configure the log level for particular modules, e.g.

"LogLevel info ssl:warn"

#

LogLevel warn

Include module configuration:

IncludeOptional mods-enabled/*.load

IncludeOptional mods-enabled/*.conf

Include list of ports to listen on

Include ports.conf

Sets the default security model of the Apache2 HTTPD server. It does

not allow access to the root filesystem outside of /usr/share and /var/www.

The former is used by web applications packaged in Debian,

the latter may be used for local directories served by the web server. If

your system is serving content from a sub-directory in /srv you must allow

access here, or in any related virtual host.

<Directory />

Options FollowSymLinks

AllowOverride None

Require all denied

</Directory>

<Directory /usr/share>

AllowOverride None

Require all granted

</Directory>

<Directory /var/www/>

Options Indexes FollowSymLinks

AllowOverride None

Require all granted

</Directory>

#<Directory /srv/>

Options Indexes FollowSymLinks

AllowOverride None

Require all granted

#</Directory>

AccessFileName: The name of the file to look for in each directory

for additional configuration directives. See also the AllowOverride

directive.

#

AccessFileName .htaccess

#

The following lines prevent .htaccess and .htpasswd files from being

viewed by Web clients.

#

<FilesMatch "^\.ht">

Require all denied

</FilesMatch>

#

The following directives define some format nicknames for use with

a CustomLog directive.

#

These deviate from the Common Log Format definitions in that they use %O

(the actual bytes sent including headers) instead of %b (the size of the

requested file), because the latter makes it impossible to detect partial

requests.

#

Note that the use of %{X-Forwarded-For}i instead of %h is not recommended.

Use mod_remoteip instead.

#

LogFormat "%v:%p %h %l %u %t \"%r\" %>s %O \"%{Referer}i\" \"%{User-Agent}i\"" vhost_combined

LogFormat "%h %l %u %t \"%r\" %>s %O \"%{Referer}i\" \"%{User-Agent}i\"" combined

LogFormat "%h %l %u %t \"%r\" %>s %O" common

LogFormat "%{Referer}i -> %U" referer

LogFormat "%{User-agent}i" agent

Include of directories ignores editors' and dpkg's backup files,

see README.Debian for details.

Include generic snippets of statements

IncludeOptional conf-enabled/*.conf

Include the virtual host configurations:

IncludeOptional sites-enabled/*.conf

└─\$ sudo nikto -h http://localhost

- Nikto v2.5.0

+ Target IP: 127.0.0.1

+ Target Hostname: localhost

+ Target Port: 80

+ Start Time: 2025-08-06 06:40:50 (GMT-4)

+ Server: Apache/2.4.65 (Debian)

+ /: The anti-clickjacking X-Frame-Options header is not present. See: <https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Frame-Options>

+ /: The X-Content-Type-Options header is not set. This could allow the user agent to render the content of the site in a different fashion to the MIME type. See: <https://www.netsparker.com/web-vulnerability-scanner/vulnerabilities/missing-content-type-header/>

+ No CGI Directories found (use '-C all' to force check all possible dirs)

+ /: Server may leak inodes via ETags, header found with file /, inode: 29cd, size: 60474c69373a0, mtime: gzip. See: <http://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2003-1418>

+ OPTIONS: Allowed HTTP Methods: OPTIONS, HEAD, GET, POST .

+ /server-status: This reveals Apache information. Comment out appropriate line in the Apache conf file or restrict access to allowed sources. See: OSVDB-561

+ 7850 requests: 0 error(s) and 5 item(s) reported on remote host

+ End Time: 2025-08-06 06:41:09 (GMT-4) (19 seconds)

+ 1 host(s) tested

└─\$ curl -I http://localhost

HTTP/1.1 200 OK

Date: Wed, 06 Aug 2025 10:42:19 GMT

Server: Apache/2.4.65 (Debian)

Last-Modified: Sun, 03 Sep 2023 13:57:34 GMT

ETag: "29cd-60474c69373a0"

Accept-Ranges: bytes

Content-Length: 10701

Vary: Accept-Encoding

Content-Type: text/html